ENGINEERING & PRODUCT BLUEPRINT

AI Personal Career Counselor

Complete Project Requirements, Technical Specifications, Data Schemas & Guided Journey Architecture

DOCUMENT VERSION	CORE ARCHITECTURE	AI REASONING ENGINE	LAST UPDATED
v2.1 (Production PRD)	React 19 + FastAPI + Supabase	Groq (GPT-OSS 120B) + MiniLM	October 2026

1. Executive Summary & Vision

The **AI Personal Career Counselor** is an intelligent guidance platform designed for college students (undergraduates, diploma holders, and fresh graduates). The application bridges the gap between passive career quizzes and static skill roadmaps by providing a two-stage holistic model:

- **Stage 1: Counselor Mode** — Answers "*Which career actually fits who I am?*" by combining psychometric profiling (Holland RIASEC 24-item test), optional PDF resume parsing, subconscious identity exploration ("Know Me" values, energy sources, anti-goal dealbreakers), and a hybrid vector matching algorithm across 1,016+ O*NET career definitions.
- **Stage 2: Daily Coach Mode** — Answers "*What specific action do I take today to get hired?*" through a gamified level map, AI-powered skill-gap analysis, dynamic weekly study plan schedules, checkpoint mastery verifications, and an interactive context-aware AI coach.

> **Core Product Principle: Choose First, Then Plan**
>
> A student must never be confronted with a generic checklist or roadmap before they have confirmed an authentic, validated career goal. The application enforces a strict, guided progression to ensure high user conviction and completion rates.

2. End-to-End Guided Student Journey

The application follows a continuous, milestone-driven progression guarded by client-side and server-side route policies:

1 Authentication & Session Management

Completed & Secured

Students sign up or authenticate via Supabase Auth (Email/Password or Google OAuth). Returning students automatically resume at their incomplete milestone rather than getting redirected prematurely.

2 Academic Onboarding & Resume Upload (`/onboarding`)

Verified Flow

Captures educational tier (Degree, Diploma, PUC/12th, SSLC/10th), institution name, course/branch, academic year, personal interests, and weekly study time availability. Optional drag-and-drop PDF resume upload extracts technical competencies and prior experience via PyMuPDF and Groq structured JSON inference.

3 Psychometric & Subconscious Assessment (`/assessment`)

Verified Flow

Comprises two tightly integrated assessment layers:

- **24-Question Holland Code (RIASEC):** Mathematically evaluates Realistic, Investigative, Artistic, Social, Enterprising, and Conventional affinity vectors (0–100%).
- **"Know Me" Subconscious Profiling:** Evaluates inner drive using the Equal-Salary test, energy source audit, secret envy compass, preferred work environment, and anti-goal dealbreakers.

4 **Hybrid Career Matching & Exploration (`/api/career-match`)** Verified Pipeline

Calculates compatibility using a deterministic multi-factor pipeline:

- **50% Semantic Vector Similarity:** Dense embeddings computed via `all-MiniLM-L6-v2` matching student profile texts against 1,016 O*NET occupations.
- **30% RIASEC Alignment:** Euclidean distance between student Holland code and career archetype codes.
- **20% Direct Skill Overlap:** Parsed resume competencies matched against target skill dictionaries.
- **Penalties:** Deductions for anti-goal dealbreakers and missing educational prerequisites.
- **AI Reranking:** Groq LLM (GPT-OSS 120B) contextual synthesis with immediate rule-based fallback if offline.

5 **Career Goal Commitment & Persistence (`career_goals`)** Task 2 Verified

The student reviews recommendations, explores in-depth salary trends and daily life via `/careers/:slug`, and confirms their chosen career goal. The selection is atomically recorded in Supabase `career_goals` (enforced by a partial unique index allowing exactly one active goal) and synced with `profiles.target_role`.

6 **Career Mode Dashboard & Daily Execution (`/dashboard` , `/today`)** Task 3 Route Guarded

The entire application transitions into personal career mode ("Good morning, future Software Engineer 🌱"). Generates the student's personalized roadmap, skill-gap analysis (`/api/skill-gap`), daily task scheduler, and streak tracker.

3. System Architecture & Tech Stack

LAYER	TECHNOLOGY & FRAMEWORK	KEY RESPONSIBILITIES
Frontend	React 19, Vite, TypeScript, Tailwind CSS v4, Framer Motion, Lucide Icons	Responsive, accessible mobile-first UI; ProtectedRoute milestone state guards; AuthContext & ProfileContext state management; optimistic UI updates.
Backend API	FastAPI (Python 3.12), Uvicorn, Pydantic v2, PyMuPDF (`fitz`)	High-performance async REST API; secure PDF document extraction; RIASEC deterministic scoring; AI proxy and caching layer.
Matching & NLP	Sentence-Transformers (`all-MiniLM-L6-v2`), PyTorch, NumPy	Dense vector embeddings for 1,016 O*NET occupations; Cosine similarity scoring; multi-factor ranking algorithms.
Generative AI	Groq LPU Cloud (`openai/gpt-oss-120b` , JSON Mode)	Structured resume JSON parsing; contextual career recommendation rationale; roadmap generation; coaching chatbot.
Database & Auth	Supabase (PostgreSQL 15), Row Level Security (RLS), Supabase Auth	User authentication; multi-tenant RLS isolation; persistent relational storage for profiles, assessments, goals, and roadmaps.

Resilience & Fallback Architecture

All AI-powered endpoints feature 100% deterministic rule-based fallbacks. If the Groq API experiences rate limits or network latency, the platform automatically produces mathematical RIASEC matches and template-based roadmap milestones without user disruption.

4. Functional Module Specifications

CODE	MODULE NAME	PHASE	FUNCTIONAL DELIVERABLES & ACCEPTANCE CRITERIA
C0	Data Foundation & RLS	Counselor	PostgreSQL tables with Row Level Security. Isolates student profiles, psychometric assessments, and career goal records.

CODE	MODULE NAME	PHASE	FUNCTIONAL DELIVERABLES & ACCEPTANCE CRITERIA
C1	"Know Me" Profile	Counselor	5 discovery sections: core values, subjects enjoyed, perceived strengths, work settings, and financial/timeline realities.
C2	AI Counselor Conversation	Counselor	Voice (Web Speech API) & text conversational counselor assessing qualitative interests across education tracks.
C3	Counselor Summary Report	Counselor	Exportable and shareable diagnostic summary: Holland code, RIASEC breakdown, top 5 matched careers, and personal trade-offs.
C4	Career Catalog & Explorer	Counselor	Searchable directory of 1,056 careers with Indian salary tiers, required competencies, daily routines, and 3-way career comparison.
C5	Career Goal Confirmation	Counselor	Explicit commitment modal saving target title, career slug, and industry field to `career_goals` before unlocking dashboard.
M0	Backend Token Verification	Coach	Supabase JWT bearer token validation on protected endpoints; strict CORS origin limits; 5 MB resume upload cap.
M1	Career Mode UI Context	Coach	Personalized global theme, field-specific color accents, time-aware greetings, and career-specific dashboard navigation.
M2	AI Roadmap Generator	Coach	Hierarchical study roadmap (4 Worlds × 3–6 Levels) tailored to career, graduation year, and study hours.
M3	NLP Skill-Gap Analysis	Coach	Compares parsed resume competencies against target career requirements into Have, Partial, and To-Learn categories.
M4	Gamified Nature Level Map	Coach	Interactive visual winding map across 4 biomes (Seed Meadow → Growing Forest → River Valley → Summit).
M5	Verification Checkpoints	Coach	Outcome-based quizzes (≥70% pass threshold) and project submission portfolios required to unlock downstream stages.
M6	Daily Planner & Today View	Coach	Transforms multi-week roadmap into 2–3 bite-sized daily tasks with automatic task rollover for missed days.
M7	Motivation Engine & Streaks	Coach	Contextual daily encouragement matching user milestones (first day, milestone reached, streak freeze, comeback).
M8	Job-Readiness Score	Coach	Deterministic readiness calculation: 60% Level Mastery + 25% Checkpoint Scores + 15% Project Portfolio.
M9	AI Coach Chatbot with Memory	Coach	Context-aware daily assistant retaining conversation summaries, open follow-ups, and proactive nudges.

5. Database Schema & Security (Supabase PostgreSQL)

All user tables enforce multi-tenant isolation via PostgreSQL Row Level Security (RLS). Users can only query and mutate records matching `auth.uid() = user_id`.

5.1 Student Profiles & Psychometric Assessments

```
-- 1. Student Academic Profiles
CREATE TABLE profiles (
  user_id        UUID PRIMARY KEY REFERENCES auth.users ON DELETE CASCADE,
  full_name      TEXT,
  college        TEXT,
  course         TEXT,
  year           TEXT,                    -- '1st year' | '2nd year' | '3rd year' | 'Graduate'
  interests      TEXT[] DEFAULT '{}',
  hours_per_week INT DEFAULT 10,
  target_role    TEXT,                    -- Synced with active career goal
  resume         JSONB,                   -- {name, email, skills[], education[], experience[]}
```

```
  created_at     TIMESTAMPTZ DEFAULT NOW(),
  updated_at     TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE profiles ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access own profile" ON profiles
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);

-- 2. Psychometric Assessments
CREATE TABLE assessments (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL REFERENCES auth.users ON DELETE CASCADE,
  answers     JSONB NOT NULL,            -- {item_id: score}
  scores      JSONB NOT NULL,            -- {R: 0-100, I: 0-100, A: 0-100, S: 0-100, E: 0-100, C: 0-100}
  code        TEXT NOT NULL,             -- Top 3-letter Holland code, e.g. 'IRC'
  matches     JSONB,                     -- Cached top matching career objects
  source      TEXT CHECK (source IN ('ai', 'rule')),
  created_at  TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE assessments ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access own assessments" ON assessments
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);
```

5.2 Confirmed Career Goals (Task 2 Implementation)

```
-- 3. Confirmed Career Goals
CREATE TABLE career_goals (
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id       UUID NOT NULL REFERENCES auth.users ON DELETE CASCADE,
  career_slug   TEXT NOT NULL,
  career_title  TEXT NOT NULL,
  field         TEXT NOT NULL,
  status        TEXT NOT NULL DEFAULT 'active' CHECK (status IN ('active', 'archived')),
  confirmed_at  TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE career_goals ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access own career goals" ON career_goals
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);

-- Enforce exactly ONE active career goal per student
CREATE UNIQUE INDEX career_goals_one_active ON career_goals(user_id) WHERE status = 'active';
```

5.3 Roadmaps and Learning Progress

```
-- 4. Roadmaps and Learning Progress
CREATE TABLE roadmaps (
  id               UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id          UUID NOT NULL REFERENCES auth.users ON DELETE CASCADE,
  goal_id          UUID NOT NULL REFERENCES career_goals ON DELETE CASCADE,
  template_slug    TEXT NOT NULL,
  template_version INT NOT NULL DEFAULT 1,
  content          JSONB NOT NULL,       -- Worlds -> Levels -> Tasks & Checkpoints
  skill_gap        JSONB,                -- {have: [], partial: [], missing: [], score: 0-100}
  created_at       TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE roadmaps ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access own roadmaps" ON roadmaps
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);

CREATE TABLE task_progress (
  user_id     UUID NOT NULL REFERENCES auth.users ON DELETE CASCADE,
  roadmap_id  UUID NOT NULL REFERENCES roadmaps ON DELETE CASCADE,
  task_id     TEXT NOT NULL,             -- '{level_id}::{index}'
  done_at     TIMESTAMPTZ DEFAULT NOW(),
  minutes     INT DEFAULT 0,
  PRIMARY KEY (roadmap_id, task_id)
);
ALTER TABLE task_progress ENABLE ROW LEVEL SECURITY;
```

```
CREATE POLICY "Users access own task progress" ON task_progress
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);
```

6. REST API Specifications & Contracts

METHOD	ENDPOINT	AUTH	REQUEST / RESPONSE CONTRACT & BEHAVIOR
`GET`	`/`	Public	Health check. Returns `{ status: "ok", ai_enabled: true, model: "gpt-oss-120b" }`.
`GET`	`/api/careers`	Public	Returns list of career taxonomy titles, slugs, and industry fields. Supports `?search=` and `?field=`.
`GET`	`/api/careers/{slug}`	Public	Detailed career profile: Indian salary tiers (Fresher/Mid/Senior), daily routine, key skills, pros/cons, and similar careers.
`POST`	`/api/upload-resume`	Bearer	Accepts `multipart/form-data` (max 5 MB PDF). Returns structured JSON: `{ name, education[], skills[], experience[] }`.
`GET`	`/api/assessment/questions`	Public	Returns 24 psychometric question objects: `{ id, text, dimension: "R"|"I"|"A"|"S"|"E"|"C" }`.
`POST`	`/api/assessment/score`	Bearer	Input: `{ answers: { [id]: score } }`. Returns normalized 0–100 scores per dimension and 3-letter Holland code.
`POST`	`/api/career-match`	Bearer	Input: RIASEC scores, student background, skills, and "Know Me" values. Computes hybrid dense vector similarity + RIASEC Euclidean distance + skill overlap. Returns top 5 careers with custom match reasons and watch-outs.
`POST`	`/api/skill-gap`	Bearer	Input: `{ resume_skills: string[], target_career_slug: string }`. Maps skills against occupational requirements into `have`, `partial`, and `missing` arrays with a job-readiness percentage.
`POST`	`/api/roadmap/generate`	Bearer	Generates or retrieves cached study roadmap for the confirmed career goal, customized by student graduation year and hours.
`POST`	`/api/counselor/chat`	Bearer	Conversational counselor agent diagnosing student situation, academic tier, energy drivers, and dealbreakers.

7. Journey Guards & Routing Safeguards (Tasks 1–4 Verification)

To guarantee that students navigate the application strictly according to their actual progression stage, the frontend utilizes centralized route guards in `ProtectedRoute.tsx`:

STUDENT STAGE	DATABASE MILESTONE STATE	ROUTING ENFORCEMENT
Unauthenticated	No active Supabase auth session	Redirect to `/login`. Retains target destination in router state.
Incomplete Onboarding	`hasCompletedOnboarding(profile) == false`	Allowed only on `/onboarding`. Attempts to visit `/assessment` or `/dashboard` redirect to `/onboarding`.
Incomplete Assessment	`onboarding complete`, but `latestAssessment == null`	Allowed on `/assessment`. Premature access to `/dashboard`, `/roadmap`, or `/today` redirects to `/assessment`.
Goal Not Confirmed	`assessment complete`, but no active `career_goals` row	Redirect to `/assessment?step=results` to force deliberate career goal selection.
Fully Committed	Active goal in `career_goals` and matching `target_role`	Unrestricted access to `/dashboard`, `/roadmap`, `/today`, `/skills`, and `/profile`.

> **Anti-Loop State Hydration**
> Route guards wait for `authLoading` and `profileLoading` to resolve before triggering redirects. This eliminates infinite redirect loops and preserves user inputs across page refreshes.

8. Non-Functional & Operational Requirements

- **Fault-Tolerant AI Fallbacks:** AI inference failures (Groq 429, timeouts, or format issues) automatically trigger deterministic RIASEC Euclidean matching and static roadmap templates. The user experience never crashes due to AI outage.
- **Data Privacy & Security:** PDF resumes are processed in memory and never persisted in plain storage. Supabase Row Level Security strictly isolates each student's data. Backend service role keys are never exposed to browser clients.
- **Performance Benchmarks:** Static catalog loading < 200ms; Deterministic scoring < 50ms; Semantic vector search < 150ms; AI reranking < 2.5s with optimistic UI indicators.
- **Cross-Device Accessibility:** Mobile-first responsive layout (A4/mobile responsive grid); Accessible WCAG AA color contrast ratios across light and dark modes.

9. Verification & Test Evidence (Tasks 1–4 Audits)

The full student journey and route guards have been tested and verified across 8 dedicated test suites:

TEST SUITE / VERIFICATION AREA	SCOPE	RESULT	VERIFIED BEHAVIOR
`onboarding-journey.test.ts`	Task 1 Navigation	Passed (100%)	Onboarding completion immediately navigates to `/assessment` without premature dashboard redirects.
`goal-confirmation.test.ts`	Task 2 Persistence	Passed (100%)	Selected career title, slug, and field write atomically to `career_goals` and synchronize with `profiles.target_role`.
`route-guards.test.ts`	Task 3 Guards	Passed (100%)	Incomplete milestones strictly redirect to respective steps. Prevents URL bypass and eliminates redirect loops.
`journey-audit.test.ts`	Task 4 Full Journey	Passed (100%)	Complete 6-step flow verified from initial registration through daily task execution.
`ai-fallback.test.ts`	Resilience	Passed (100%)	Groq outage simulation confirmed graceful degradation to deterministic scoring without UI errors.

10. Implementation Status & Product Success Metrics

100%	1,016	< 150ms	0%
PHASE 1 COMPLETE	O*NET CAREERS	VECTOR MATCH SPEED	ROUTE LEAK RATE

Phase 2 Target Key Results

- **Assessment Completion:** ≥ 80% of registered students complete onboarding and RIASEC profiling.
- **Goal Conviction:** ≥ 95% of students explicitly confirm a career goal before accessing daily roadmap features.
- **Daily Habit Formation:** ≥ 70% weekly active retention via dynamic rollover tasks and streak motivation.

Architecture Review & Sign-Off: All Phase 1 modules (C0–C5) and student journey route guards (Tasks 1–4) are verified and in production readiness. Phase 2 (Coach modules M2–M9) will branch from the confirmed `career_goals` schema.